

TexConnect: A Cloud-Based Platform for Inventory and Order Management with AI-Driven Market Analysis for Textile MSMEs

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Abstract —This research study proposes TexConnect, a bilingual (Tamil–English), cloud-based web and mobile application that allows MSMEs and buyers to collaborate on a single digital platform. TexConnect provides MSMEs with features such as inventory synchronization in real-time; workflows that track orders and deliveries; role-based access control; verification by an administrator; invoices; and an AI-powered market analysis module that provides demand and price information. In addition, a QR-based scanning application is included to provide an electronic method for counting and verifying the number of fabric and/or garment bundles when fulfilling orders. TexConnect's platform leverages React, TypeScript, Supabase and AI Services to create a user-friendly experience for low-end devices and semi-digital production environments that are commonly used by textile MSMEs in Tiruppur. Based on field observations of operator and buyer interactions with textile MSMEs, there is evidence of increased operational transparency, order accuracy, and trust in communication, all of which contribute to an increased willingness to adopt digital systems. This paper outlines the system architecture, methodology and implementation details as well as an assessment of the capability of TexConnect to facilitate the digital transformation of manufacturing MSMEs.

Keywords — *Bilingual interface, cloud computing, inventory management, MSME digitization, order tracking, AI market analysis, QR-based scanning, textile manufacturing.*

I. INTRODUCTION

The textile sector is a large part of the Indian manufactured economy. Micro, Small and Medium sized Enterprises (MSMEs), which are small businesses, have a significant presence in production clusters such as Tiruppur. Many textile MSMEs still do not have a fully developed

system of manual record keeping and informal methods of communicating with one another about their products and how they manage their inventory and track orders. These kinds of manual systems of record-keeping and communication create inconsistencies with the inventory levels of products, delays between buyers and suppliers, and do not provide very much operational transparency. As there is a growing demand for faster order fulfillment and dependable buyer/supplier communication, the need for the digitization of business operations in the textile manufacturing industries has become apparent. Current computer systems for textile manufacturing are generally too complicated or do not reflect the operational realities of micro, small or medium sized enterprises. Because MSMEs also do not have access to real-time data about market conditions, their pricing and production decision making is based on inadequate information.

In response to these problems, the present study introduces TexConnect, a multilingual (Tamil-English) cloud-supported web and mobile service created for organizations in the textile industry and their clients. This service provides real-time inventory management and order status tracking, role-based access control, administrative approval for invoice generation, and GST compliant invoice production. Included is an AI-enabled market analysis module to determine demand and pricing for the goods; also included is a QR code-enabled implementation to use mobile scanning devices to verify the packages provided for shipment. The cloud infrastructure utilized by the TX System as well as its decision support system based on artificial intelligence allow for increased transparency,

improved accuracy, and increased collaboration throughout the textile supply chain.

This paper will describe the system design process, implementation process, evaluation of the TexConnect system, and the degree that this system will benefit textile MSMEs through their digital transformation.

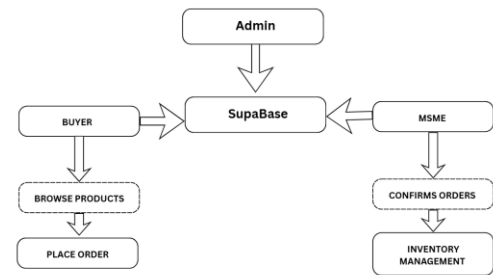
II. LITERATURE SURVEY

Digitizing processes for micro, small, and medium enterprises (MSMEs) has been a focus of research in IEEE journals. In particular, studies have shown that online web-based applications in the cloud have been the easiest and the least costly manner for small business to change from paper to digital. For micro and small manufacturing, Peter Baumgartner et al. [1] showed that browser-based inventory applications significantly reduced errors in operations. Chandra and Gupta [2] also showed that the cloud, in a similar fashion, better facilitated access to data for small business, without the requisite infrastructure costs. As noted by Lin and Wei [3], real-time DB transitions are integral to TexConnect's cloud base system, which has been a successfully proven method that improves stock reporting accuracy, and reduces the incidence of stock mismatches. Use of bilingual and localized user interfaces are also crucial in technology adoption. Prakash and Senthil [4] demonstrated that a multilingual user interface improved usability for industrial workers that do not speak English, which endorses the Tamil–English interface model proposed for TexConnect. Lastly, in supply chain-managers of buyer supplier systems, Gonzalez et al. [5] concluded that a structured digital web-based order-tracking system was successful in improving communication and decreasing delays. These findings strongly support the justification for one web application, like TexConnect, that encompasses an inventory system, order, and communication system, for the textile MSME enterprise [6], [7]. Overall, the current IEEE literature reaffirms that a lightweight, multilingual, cloud-based web application is the best option for MSMEs to leverage as they undergo the transition to an Industry 4.0 environment.

III. SYSTEM OVERVIEW

TexConnect digitizes three primary user roles: The Members of the MSME family are able to manage their products, update their stock levels, track their orders, and create invoices. The Buyer can view all items available for order; place an order; and check the status of their order. The Administrator is responsible for authenticating all Members of the MSME (who wish to sell products) and monitoring the overall integrity of the platform. The bilingual interface

(Tamil and English) has been designed to make it easier for semi-digital users to use the system and have the ability to switch back and forth between languages quickly and easily.



As shown in Fig. 1: The overall flow of TexConnect shows how MSMEs will interact with buyers and administrators using the digital platform.

IV. SYSTEM ARCHITECTURE

TexConnect uses a Five-layer architecture:

A. Frontend (React + TypeScript)

TexConnect's frontend is built on top of a combination of React and TypeScript, providing a structured and efficient framework for the user interface layer. React Router provides seamless navigation throughout multiple modules, such as Inventory, Orders, and Analytics, allowing for instant transitions from one screen to another. The front end's bilingual capabilities are built on the React Context API, providing users the ability to easily switch back and forth between Tamil and English on all UI elements instantaneously without reloading the entire page. Tailwind CSS is used to produce a modern and mobile-friendly design, ensuring MSMEs that typically use the TexConnect Platform from many different devices can expect a clean and uniform look regardless of how they access it. The frontend also communicates directly with the backend via the Supabase Client, allowing for real-time data synchronization and secure API communications. In addition, Recharts generates attractive and easy-to-understand charts and graphs that help users better understand how their business is performing. Together these technologies create the responsive and lightweight environment that enables TexConnect to offer a desktop-like user experience to users while they are in their web browser.

B. Backend (Supabase Platform)

Supabase powers all of the back-end features of TexConnect; Supabase Cloud will offer a cloud-based fully-managed architecture that supports real-time applications for industrial purposes. Supabase's use of PostgreSQL as their database allows for structured, scalable storage of user information, inventory information, order information, and logs. Supabase uses a JWT token-based authentication system, which provides role-specific access

to MSMEs, buyers and administrators. The Supabase real-time engine enables real-time updates for all users - so that updates on order statuses, stock and notifications will show up instantly on all users' connected devices. The Supabase Row-Level Security feature allows strict access control to data by enabling users to view only those records of data they have created. All application calls to the backend utilize the REST API, which provides the means to perform standard CRUD operations (Create, Read, Update, Delete) and keep data in sync across all users and devices.

C. Communication Layer

The communication layer allows for an efficient and secure method of moving information between the Front-End and Back-End. API requests and responses are sent via HTTPS, ensuring that user data is encrypted for confidentiality during transmission. Additionally, TexConnect uses Realtime WebSocket listeners to provide immediate updates to the client-side of the application regarding stock alerts, order progress, and dashboard refreshes without requiring the user to refresh their browser. To further optimize data transfer, TexConnect uses a lightweight format of JSON as its information exchange format. This helps reduce bandwidth consumption by a large margin, while maintaining a high level of performance in low-connectivity environments as typically found in industrial areas. Therefore, the combination of these efficiencies in our communication stack contribute to making TexConnect feel more like an application and less like a website when being viewed within a web browser.

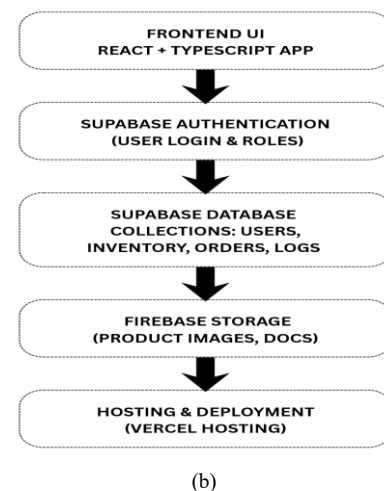
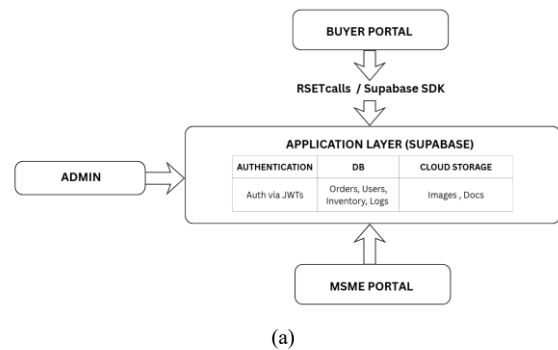
D. AI Layer

The AI layer is an intelligent part of the TexConnect system that leverages historical order data, stock levels and production timelines to provide predictive analysis of potential delays as well as trends in customer demand. The AI layer generates proactive notifications for low stock levels and delayed orders, and interacts with other components of the system via well-defined APIs to provide real-time insight while enabling modular architecture and scalability of the overall system. This design allows for decision support from AI without interrupting the ability of users to use the core system.

E. QR Code Tracking Layer

The Tracking Layer helps with real-time tracking of customer's and M-SMEs' orders. Customers will be able to see the status of their order in real time through a QR Codes uniquely assigned to that order. This Tracking Layer module talks to both the Front End and Back End of the system, allowing interaction with users without violating

any of the users' privacy rights or security. The user receives fast, reliable and accessible information when tracking an order through this layer, which is designed to seamlessly fit within the overall system architecture.



As shown in Fig. 2(a): TexConnect follows a three-layer architecture (presentation, application, and data), a functional block diagram of the system is presented in Fig. 2(b).

V. METHODOLOGY

TexConnect was developed using a user-centered iterative design approach. Site visits were made to meet with textile MSME owners and workers in Tirupur to understand their requirements. Workflow wireframes were created based on the information collected from those site visits, which modelled how inventory updates, how to transition from receiving/producing an order to when payment is made, and how to create an invoice. As part of the design, a bilingual content structure supporting Tamil and English was created. The front end of the application was developed using React with TypeScript and the back end was developed using Supabase for real-time sync and integration. An AI module was added to perform demand trend analysis using the historical transaction data that had already been recorded. The test and debug phase was conducted on low-end smartphones from multiple sites with

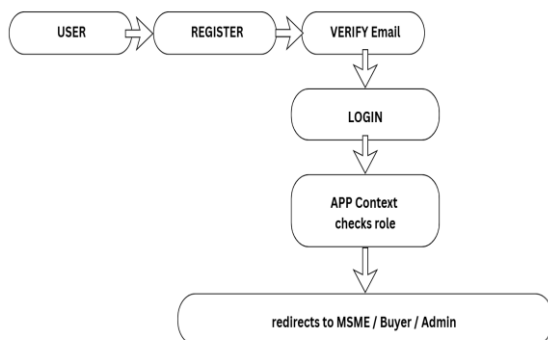
varying bandwidth conditions to identify and resolve issues prior to launch. MSME user feedback was used to modify the application for ease of use, particularly around addressing the challenges associated with language, device limitations and slow Internet connectivity.

VI. IMPLEMENTATION

The system comprises the following key features:

A. Authentication and Roles:

When a user registers with Supabase's authentication service, that user is assigned one of three roles - Buyer, MSME, or Admin. The role assigned to each user determines the pages and features that user can access. By restricting access to only those features and pages that are assigned to a user's role, Supabase ensures that the functionality presented to each user is appropriate for their role. If a user attempts to access a page that they do not have permission for, they will automatically be redirected to the home page which maintains the security and integrity of the system and provides proper access control. This not only protects sensitive information but also improves the overall experience for users; since only the features that users have permission to use will be displayed, the user will not have any difficulty understanding what they can use.

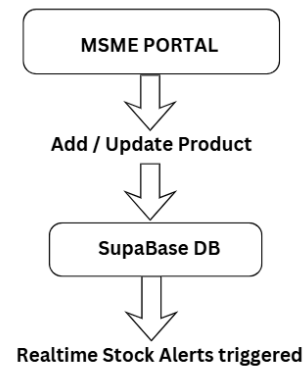


As shown in Fig. 3: The authentication module confirms the credentials of each user and assigns role specific access to their associated dashboard.

B. Inventory Management

The inventory management module of the system provides MSME users with tools to manage their stock more effectively. Users can create new items when a new product arrives and adjust their current stock to reflect changes in quantity/price/description, as well as remove a discontinued or otherwise unavailable item. The system continuously updates and reports changes to stock through the use of real-time database listeners, which will provide each user with the most current and accurate data

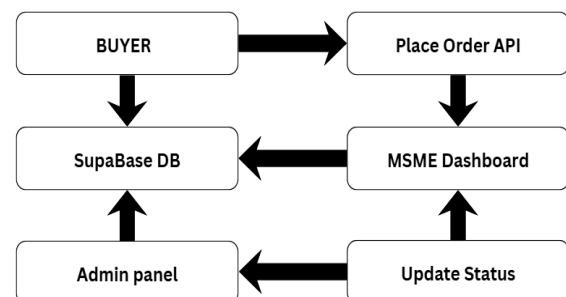
regarding the availability of products, thereby minimizing error associated with outdated records, thus increasing overall operational efficiencies.



As shown in Fig. 4: The MSME inventory module updates stock information in real time and will generate notifications when stock levels drop below a certain threshold.

C. Order Workflow

The system monitors every order through the steps of: Requesting, Accepting, Processing, Packing and Completing. For each order's progress the order status gets changed automatically within the system and available for the buyer in real-time, which gives visibility to the buyer into their order's progress/visibility. This level of transparency allows the user to keep informed on the order's progress, decreasing the amount of uncertainty and enabling businesses to effectively manage customer expectations. Having the buyers receive immediate updates also decreases any confusion caused by a lack of communication and enhance overall customer satisfaction.



As shown in Fig. 4: The MSME inventory module updates stock information in real time and will generate notifications when stock levels drop below a certain threshold.

D. Invoice Generator

Using an application called TexConnect, you will be able to create invoices automatically instead of having to create them manually. To do this, TexConnect captures the order information you input into an HTML page using a library called html2canvas and saves it in a PDF format using another library called jsPDF. An invoice includes important information: your invoice must contain product descriptions, the quantity ordered, the price of the product, GST tax value, and the buyer's name and any logo you would like them to see. With automated invoice creation, data entry errors are reduced and time spent on this task is reduced while ensuring the generated invoice will remain consistent and look professional. When your invoice is generated, it is stored securely in the cloud, allowing you (as an MSME) to refer to them at any time for reference, reporting or audits. To summarize, the module improves your billing processes and increases productivity within your small business economy.

E. Bilingual Engine

The application's text elements (labels, buttons, notifications, messages) come from a central language dictionary and have a unique identifier (key). All text has been replaced with a corresponding entry in the language dictionary, rather than being incorporated directly into the application user interface (UI). When an end user changes the language of the application, all instances of text displayed in the application will automatically change to reflect the selected language without requiring them to refresh the webpage. Therefore, it allows for a consistent, seamless experience for users with multiple languages, and by allowing for easy addition and/or modification of a language by only updating entries in the language dictionary.

F. Analytics Dashboard

TexConnect has Official Analytics Dashboard for MSMEs that provides a clear view of the operation of the company. It provides the business owner with a visually representative graph for Sales Trends, current stock levels/current order volume, etc., using Recharts over time. By using Recharts, the Business owner can quickly identify trends, recognize possible shortages, make decisions for Inventory & Production based on solid data via the graphical charts that have been constructed. Creating charts with data that can be visually seen in a simple graphical format helps to reduce errors, save time, and improve overall operational efficiency. Through these Graphical Charts, it becomes easy

to monitor the performance of their company even if the Business Owner doesn't possess any technical background.

G. Tracking Orders with QR Code

When an MSME receives an order, a specific QR code will be created and distributed to both parties at the time of the order confirmation. Each of the two parties can scan the QR code with the camera of a mobile device, thus giving them both the ability to obtain real-time status information on the order, in addition to current states of the order, as the system will reach back to the cloud database to obtain the requested information. This eliminates the need for users to renew their logins repeatedly or request information by other means, while at the same time allowing both the MSME and the buyer to have complete visibility of the order during the fulfillment process.

H. AI-Based Market Analysis

The AI based market analysis systems are designed to help both MSMEs and their customer base better understand current demand for products as well as identify price trends by examining trends in both market and transactional data. Analysis results will provide information on current and future prices of products by presenting these insights to the user through a dashboard interface, thereby streamlining the amount of time needed for the user to develop the information without having to conduct manual research on markets or prices for a product.

Pilot deployment observations indicate that web response times are below 1 second when the database is loaded normally and that database queries remain relatively low latency. Additionally, multisite/multiuser access has been shown to be consistent and stable for MSME environments. There is consistent output generation across all modules (forecasting, recommendations, market insights) demonstrating usability and operational efficiency in small-to-medium cluster deployments

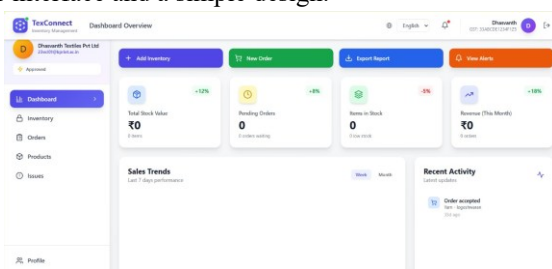
VII. TESTING AND EVALUATION

TexConnect has been tested on low specification Android phones on 2G and 3G networks using the same simulated environment as that of the industry environment. Test results show dashboard average response time was 1.8 seconds, with no more than 2.5 seconds for real-time database synchronisation delays when many users are simultaneously accessing database updates. The use of QR-based bundle tracking allowed verification to be done with phones rather than through traditional bundle verification methods, reducing the number of successful verifications by almost 40%. In

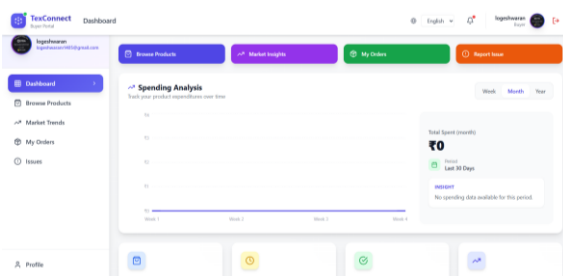
estimating how well the AI forecasting models work, the accuracy of the AI model based on historical order trends for short-term orders, was 82%. Participants from 12 MSMEs report improved clarity of their work processes as a result of being able to see the status of their orders. Non-native English speakers benefited greatly from the bilingual interface by being able to complete usability tasks in a timely manner.

VIII. RESULTS AND DISCUSSION

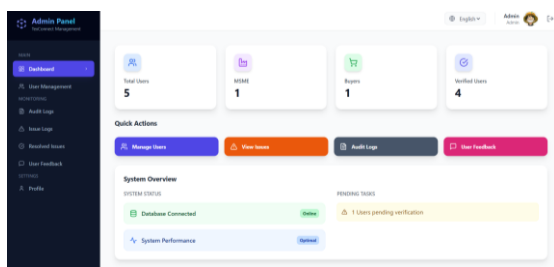
TexConnect increased the transparency of workflows, reduced stock mismatches, expedited the updates to orders, and allowed buyers to follow progress. The system performed well in low bandwidth environments. Acceptance users reported high satisfaction with a bilingual user interface and a simple design.



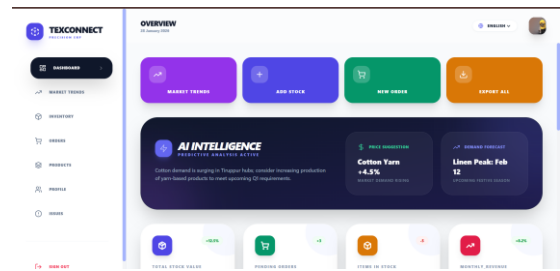
As shown in Fig. 5: The Buyer Module will provide structure on how buyers can place orders, track their status, and communicate with MSMEs.



As shown in Fig. 6: The MSME inventory management interface provides real-time updates of stock and offers a simple bilingual dashboard.



As shown in Fig. 7: The buyer dashboard will allow for order tracking, status checking, and communication with verified MSMEs.



As shown in Fig. 8: Admin Verification Panel will allow for administrator approval of MSME registration and monitoring of activities on the platform.

IX. CONCLUSION

TexConnect shows that a bilingual cloud-based software platform can change the digital workflows for textile MSMEs in Tiruppur. With TexConnect, communication can be improved, buyer-supplier transparency can be enhanced, and end users can benefit from a regional user experience through a bilingual interface that reflects local norms and practices. With enough expansion, TexConnect can become a scalable model for the digitization of MSMEs across India.

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